

## 2.1 INTRODUCTION

The internal conditions of an organism is referred its internal environment it includes  $H_2O$  quantity, different solutes, temperature, etc. For proper metabolic functions body requires all these conditions at particular level. So homeostasis is set of metabolism which maintain internal environment of an organism within suitable limits. Why should the internal environment remain constant? Let's take the example of temperature, the temperature of external environment continuously change during the day but the enzymes work within a certain range of temperature therefore, the living organisms must keep their internal temperature within this range. Organisms maintain internal condition by feedback mechanism. How body maintain homeostasis by feedback mechanism.

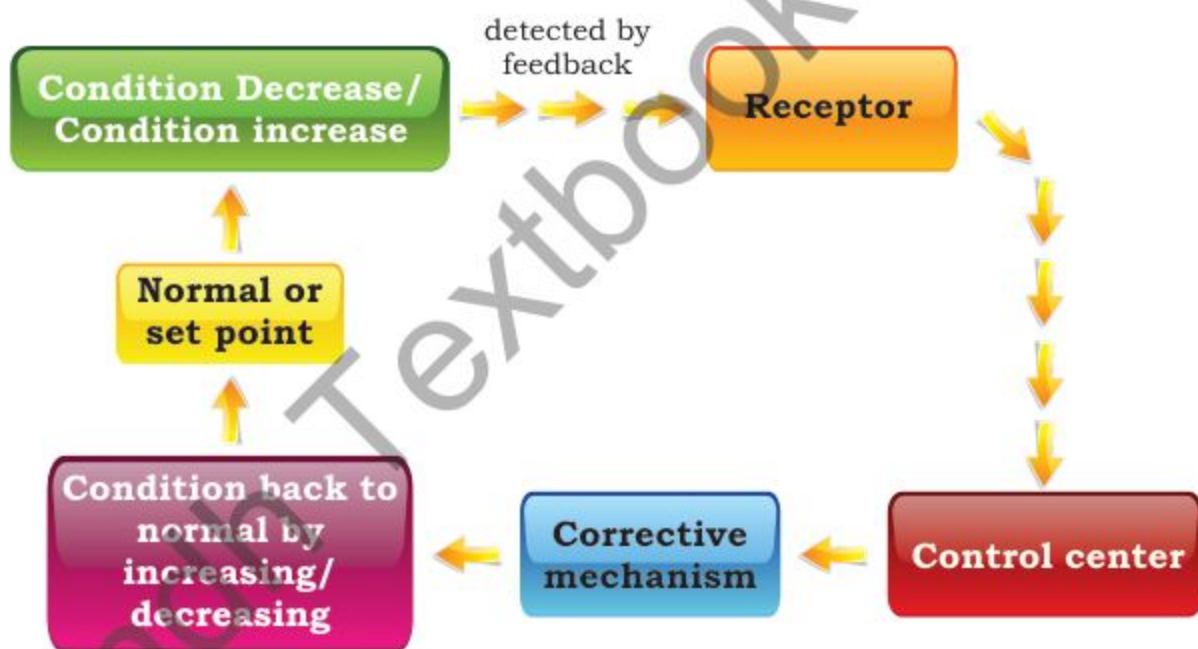


Fig. 2.1 Homeostasis by feed-back

## 2.2 ADAPTATION OF PLANT FOR DIFFERENT INTERNAL CONDITION

There are three main aspect of homeostasis

### (i) Osmoregulation:

It is the maintenance of internal water and salt conditions by osmosis.

## (ii) **Thermoregulation:**

The maintenance of temperature within suitable limits where enzymes can work optimally.

## iii) **Excretion:**

The process where metabolic toxic waste or excess metabolic substances from body i.e  $\text{NH}_3$ , urea or uric acid, gums, latex etc.

### 2.2.1 **Excretion or Storage of $\text{CO}_2$**

At daytime plant perform photosynthesis in green cells and respiration in all living cells. The  $\text{CO}_2$  produced in respiration utilized in photosynthesis. When rate of photosynthesis will be higher than respiration the plant gets extra  $\text{CO}_2$  from air and release extra  $\text{O}_2$  in air through stomata. At night plant only perform respiration only  $\text{CO}_2$  is produced which is removed by the process of diffusion through body surface. The green parts perform these gases exchange through stomata while non green parts perform this gaseous exchange through body surface.

### 2.2.2 **Removal of Extra Water**

The plant store large amount of water this water can be removed from plant in two ways i.e.

#### (a) **Transpiration**

#### (b) **Guttation**

Transpiration is the removal of water in the form of vapours from aerial part of plant. It occurs only at day time.

Guttation is the removal of water in the form of liquid from the margin of leaves through special pores, hydathodes. It only occurs at night when water pressure is high in leaves and low temperature environment is present.

Plants modify their leaves size, structure and structure of stomata to control the rate of transpiration.



**Fig. 2.2 Showing Guttation in Leaf**



Plants produce some secondary products like latex, resin and gum. These secondary products are insoluble, harmless compounds. Some plants produce special types of gums for example Neem or keeker etc. The extra amount of these are removed from special pores called **lenticels**. The coniferous plants produce resins like material while the rubber plants produce latex which remove from scare like openings. Some of these carnivorous plants and okra produce mucilaginous material to capture insects.

### 2.2.3. Osmotic adjustment in plants

The plants grow in different conditions of water and salts, on the basis of water and salt quantity there are four type of plants

- (a) Hydrophytes
- (b) Halophytes
- (c) Mesophytes
- (d) Xerophytes

#### (a) **Hydrophytes (hydro = water; phyta = plants)**

The plant which grow in fresh water; they live completely or partially in fresh water so called totally or partially submerged plants. They adapt themselves for removal of excess water which can enter in this condition.

- These plants do not contain roots or have poorly developed roots.
- They have broad leaves if partially submerged and have stomata at upper epidermis e.g water lilly.
- They may have thin and spongy tissues in leaves and stem in totally submerged plant e.g Hydrilla.



**Fig. 2.3 Hydrophyte**

#### (b) **Halophytes (Halos= salt)**

They grow in sea marshes or in saltish water. In salty condition water moves outside the cell which is not suitable for plants. To move water from outside to inside the plant develop following characters.



- Plants develop salt glands where plant store salts by taking it through active transport.
- Plants oppose salt to move outside from vacuole.
- Some salt accumulated at surface of leaf which attracts water from air



**Fig. 2.4 Halophyte**

**(c) Mesophytes:**

The plant grow in moderate water containing soil they will develop following characters.

- They have developed root system which do not grow very deep.
- They have moderate sized leaves.



**Fig. 2.5 Mesophyte**

**(d) Xerophytes:**

Plants grow in soil of low water quantity. They grow in desert or steep slopes or at high altitude. To conserve water and absorb proper amount of  $H_2O$  they develop following characters.

- They have vertically growing deep root system to absorb proper amount of water.
- They possess thick waxy cuticles over epidermis to conserve water.
- They have short sized leaves or leaves are modified into spines to reduce loss of water by reducing the number of stomata.
- Some xerophytes has special parenchyma cells in stem, where they store water, this makes the stem soft, wet and juicy called succulent organs e.g. cacti.



**Fig. 2.6 Xerophyte**



## 2.3 HOMEOSTASIS IN ANIMAL

### Osmoregulation in Animals

Like plants, animals also live in aquatic and terrestrial habitat. According to their environment their cells require more critical balance of water and solutes. Water continuously leaves and enters the cells with solutes to keep the water and solute in constant quantity which are required for smooth metabolic functions.

#### Osmoregulation in aquatic environment

The aquatic conditions are classified on the basis of the concentration of salt present in it. The water which contains very low amount of salt called fresh water and the water contains high salt called marine water. Animal osmoregulate differently in both waters.

##### i. Osmoregulation in fresh water

Fresh water animals have hypertonic conditions inside their body or cells so they always facing the problem of flooding of  $H_2O$  and loss of salts.

We can classify further these animals in two groups.

| Unicellular                                                                                                                           | Multicellular                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Pump out excess water by contractile vacuole</li><li>• e.g. Amoeba, Paramecium etc.</li></ul> | <ul style="list-style-type: none"><li>• Pump out excess water by producing dilute urine.</li><li>• Loss of salt is compensated by active uptake of salt by gills and skin as well as use of salt containing food.</li></ul> |

##### ii. Osmoregulation in marine animals

Usually marine animals have hypotonic conditions (low salt) inside the body but some marine animals develop hypertonic (high salt) or isotonic (same salt condition) by metabolism.



| Bony fish                                                                                                                                                                                                            | Cartilaginous fish                                                                                                                                         | Osmoconformer                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Have low salt inside the body</li> <li>• Actively get sea water and have salt glands to increase the salt and desalination</li> <li>• Produce concentrated urine</li> </ul> | <ul style="list-style-type: none"> <li>• Have high salt by storing urea inside</li> <li>• Eat food which contain nitrogenous compound i.e. meat</li> </ul> | <ul style="list-style-type: none"> <li>• Have equal amount of salt.</li> <li>• These animals do not require any activity to adjust their internal osmotic condition. i.e. unicellular.</li> </ul> |

### iii. Osmoregulation in terrestrial condition

Terrestrial conditions are harsh for living organism because the direct contact of heat to body causes loss of water which leads to dehydration, major problem for terrestrial life. Only arthropods, some molluscs reptiles, birds and mammals can survive in this habitat because:

- Their bodies are covered by exoskeleton or thick skin, which prevent loss of water.
- They conserve water by reabsorption in kidneys and rectum.
- Some of them can produce water from fats catabolism with the help of peroxysomes i.e. camel, kangaroos.
- Continuously drinking of water or using liquid food.

### Excretion:

During metabolism living organisms catabolize protein and other nitrogen containing compounds which produce some toxic nitrogenous compound. These toxic compounds are mainly  $\text{NH}_3$  or urea or uric acid generally called nitrogenous waste. If these compound retain in the body and accumulate, they can damage the cells or organs therefore they must be removed from the body. The removal of these nitrogenous metabolic waste is called excretion.

On the other hand plants metabolism is different from animal. Plants are autotrophs, initially they produce carbohydrates as primary products. Carbohydrate is catabolized to produce  $\text{CO}_2$  and  $\text{H}_2\text{O}$ . The  $\text{CO}_2$



reutilized in photosynthesis and  $H_2O$  is not a toxic compound. As autotrophs they synthesize variety of compounds, so the waste products of one reaction are utilized in other metabolic reactions as reactant and consumed.

### Excretion in animals:

The animal cells produce their nitrogenous waste during metabolism and removed them either in tissue fluid or in blood. So the animals develop some organs to filter the tissue fluid or blood. These organs are called excretory organs.

| Name of animal             | Excretory organs                              | Excretory compound       | Source         |
|----------------------------|-----------------------------------------------|--------------------------|----------------|
| Planaria (Platyhelminthes) | Flame cells (Protonephridia), Excretory ducts | Dilute urine             | Tissue fluid   |
| Earthworm (Annelids)       | Metanephridia                                 | Dilute urine             | Coelomic fluid |
| Cockroach (Arthropod)      | Malpighian tubules                            | Uric acid pellets        | Haemolymph     |
| Vertebrate                 | Kidneys                                       | $NH_3$ , Urea, Uric acid | Blood          |

## 2.4 HOMEOSTASIS IN MAN

Humans have well developed homeostasis systems. The main organs which involved in homeostasis are

- (i) Skin
- (ii) Lungs
- (iii) Kidneys

### (i) Skin:

The skin is considered as the largest organ of the body, basically functions as a protective organ as the first line of defense but it also works efficiently as a homeostatic organ by maintaining temperature, water and salt.

## (ii) Lungs:

They maintain levels of  $O_2$  and  $CO_2$  in the blood, body fluid and cells. Maintenance of  $O_2$  and  $CO_2$  level, maintain rate of respiration and continuous flow of energy.

## (iii) Kidneys:

Kidneys are called filters of the body fluids, they maintain internal water by removing excessive water, also maintain urea, uric acids, creatinine and other waste by excreting them through urine

### 2.4.1. Structure of Human Skin

Human skin consists of three layers called epidermis, dermis and hypodermis. The outer layer of skin is epidermis, made up of flat, dead cells containing keratin protein. This layer does not contain blood vessels. It is impermeable to water and prevent water loss from the body as well as work as protective layer by preventing entry to microorganisms.

Dermis is the layer present between epidermis and hypodermis, it contains many different structures i.e. nerves ending receptors to detect temperature change, pain, pressure etc. The dermis also contains sweat glands which secrete sweat on the surface to maintain temperature and also secrete urea, water and salt.

A network of arterioles are also present in the form of network, which are involved in temperature regulation. The dermis also contain hair follicle and sebaceous glands which secrete oily sebum.

Hypodermis is the inner most layer of skin containing fats which act as insulation against loss of heat. It also stores energy.

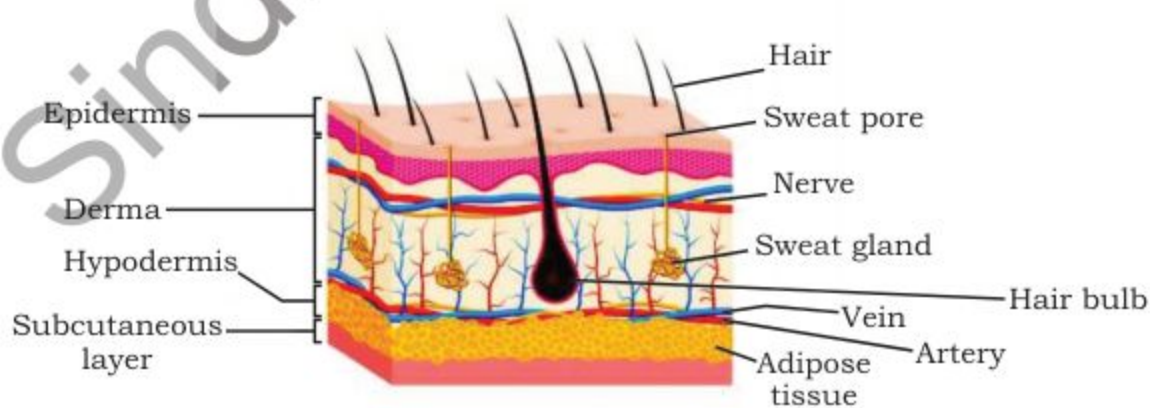


Fig. 2.7 LS of human skin



### 2.4.2. Role of Skin in Regulating Body Temperature

The skin is the organ which help in regulating body temperature when the receptor in skin detects change in body temperature from set point ( set point of human is  $37^{\circ}\text{C}$ ) i.e increase or decrease, and then Receptor send nerve impulse to the brain. It occurs by feedback mechanism to correct the temperature.

#### If Body Temperature Start Rise

**(i) Production of sweat:**

The sweat gland starts to produce and secrete sweat. The sweat accumulates at the surface of skin which evaporates with heat energy so the body feels cooling.

**(ii) Laying down of hairs:**

In hot condition, muscles which are attached with hair relax. It allows the hair to lie flat against surface of the skin

**(iii) Vasodilation:**

Arterioles found in the form of network in dermis, dilate (become wide) which increase the flow of blood, as well as it brings the blood vessels near the surface of skin which allows more heat loss. This process of vessel dilation is called vasodilation.

#### In cold condition when body temperature starts decreasing

**(i) Erection of hairs:**

The muscles contract pulling the hairs upright and trapping a layer of insulating air next to skin. Now it is not very much effective in human.

**(ii) Vasocontraction:**

Narrowing of blood arterioles of dermis occurs which reduces the blood flow in capillaries of skin so less heat is lost.

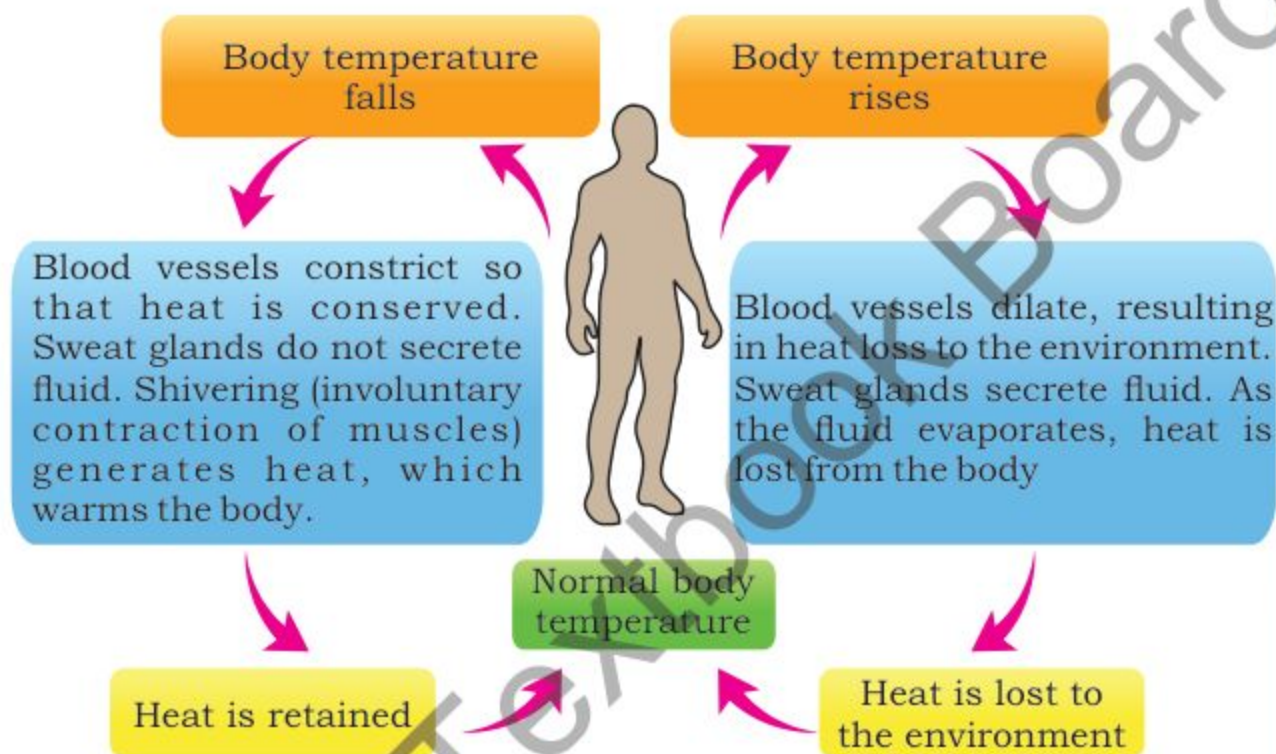
**(iii) Decrease in sweat production:**

The sweat gland stops to produce and secrete sweat, so it prevent from energy loss.



#### (iv) Increase in metabolic rate:

In cold conditions the rate of metabolism in the organs increases generating more heat which is distributed around the body in the blood stream. It prevents loss through the adipose tissue in hypodermis which work as an insulation layer.

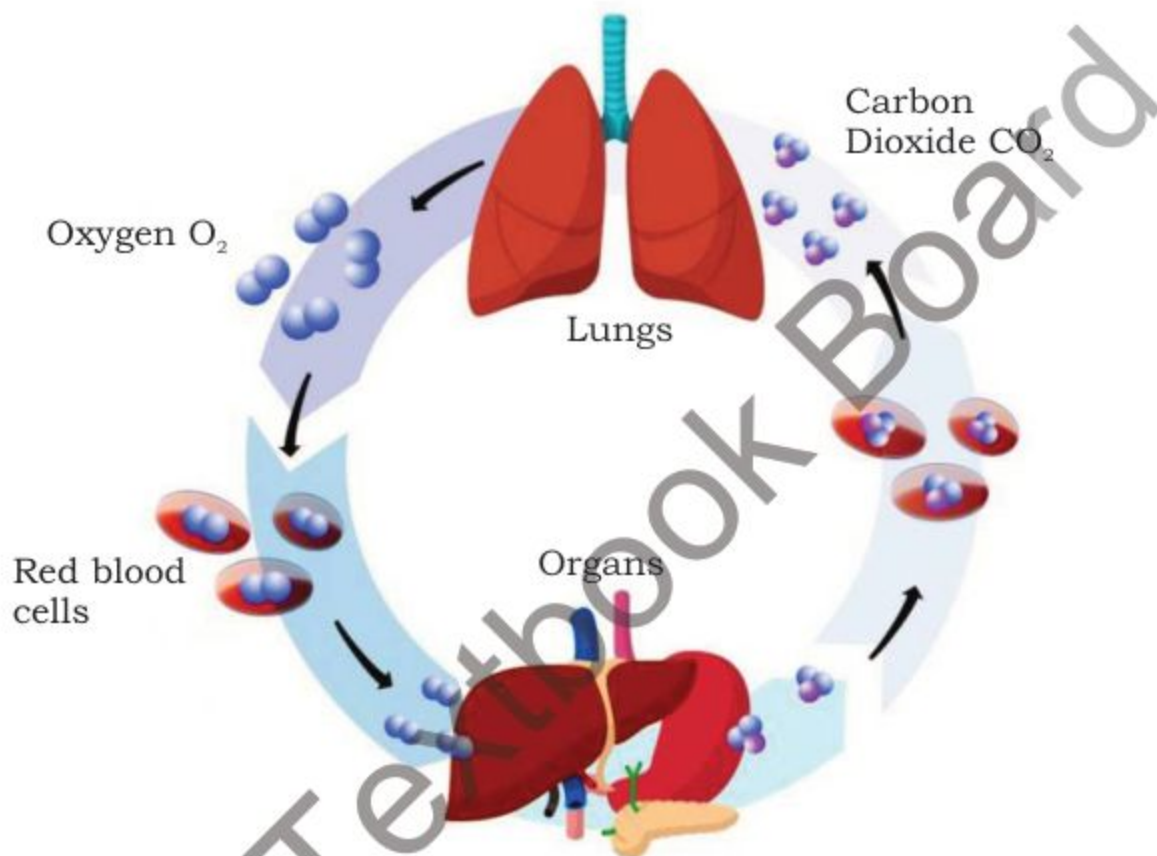


**Fig. 2.8 Thermoregulation in Man**

#### 2.4.3. Role of lungs to keep the $\text{CO}_2$ concentration low to certain level

Tissue/cells produce a large amount of  $\text{CO}_2$  during aerobic respiration. As blood passes through tissues via blood capillaries, this  $\text{CO}_2$  diffuses into the blood, where it reacts with water form carbonic acid. This reaction takes place by an enzyme called carbonic anhydrase present in R.B.C. The carbonic acid dissociate into  $\text{H}^+$  and bicarbonate  $\text{H}^+\text{CO}_3^-$  ions. The level of  $\text{H}^+$  in blood is continuously monitored by special detectors (receptor) carotid bodies and aortic bodies. Most of the bicarbonate ions diffuse out from R.B.C to blood plasma. A small amount of  $\text{CO}_2$  is also carried and dissolved in R.B.C when the blood reaches lungs these bicarbonate ions diffuse back into RBC where again converted into carbonic acid then into  $\text{CO}_2$ . The  $\text{CO}_2$  diffuses out of the blood capillaries and into alveoli, where from it is expelled out when breathing out.

If the  $\text{CO}_2$  level increases in blood, pH of blood start increasing so that the receptor sends a message to the control centre which ultimately increases the breathing rate to expel out the  $\text{CO}_2$  efficiently.



**Fig. 2.9 Role of lungs in keeping low  $\text{CO}_2$  and high  $\text{O}_2$  level in blood**

#### 2.4.4. Role of kidney in controlling blood composition

Blood is the fluid having cells. In plasma, it contains a high amount of  $\text{H}_2\text{O}$  and some solutes like  $\text{Na}^+$ ,  $\text{Cl}^-$ ,  $\text{Ca}^{++}$ ,  $\text{K}^+$  etc. with nitrogenous waste. Liver continuously produces urea and  $\text{NH}_3$  by breaking amino acid, we continuously take different solute ions in our food like  $\text{Na}^+$ ,  $\text{Ca}^{++}$ ,  $\text{K}^+$  etc. the concentration of  $\text{H}_2\text{O}$ , solute and nitrogenous waste are maintained by kidney through process of filtration and reabsorption, which we will study in next topics.

### 2.5 URINARY SYSTEM IN MAN

The urinary system of human is consist of

- A pair of kidney



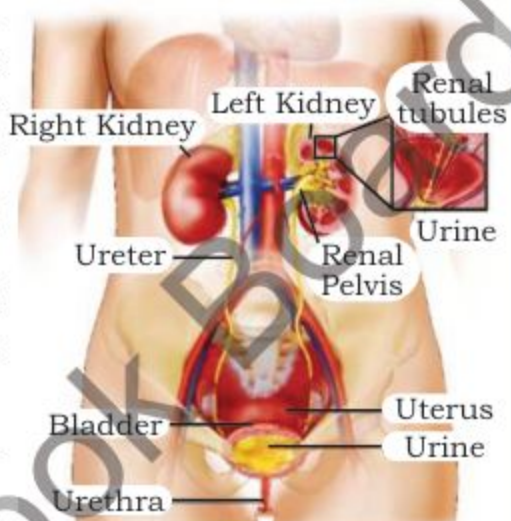
- A pair of ureters
- A urinary bladder
- A urethra

Kidneys are reddish-brown bean shaped organs, situated at the dorsal side of the abdominal cavity on either side of the vertebral column. The kidneys lie above the waistline. Each kidney has an area in the center of concave surface which faces the vertebral column; this area is called hilus. The renal artery, renal vein, nerve and ureter are connected to each kidney at the hilus.

The ureter is a narrow tube which connects the kidney to the urinary bladder. Urine passes through ureter to the urinary bladder.

The urinary bladder is a thin walled muscular bag situated towards the bottom of abdominal cavity in front of the rectum which stores urine.

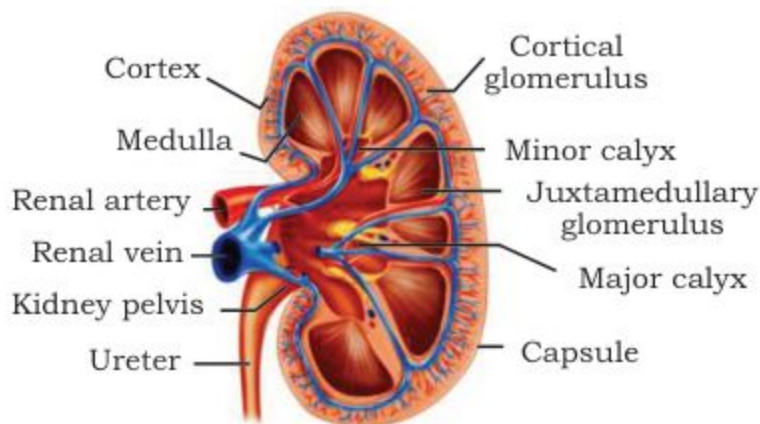
The urethra is a tube which comes out from the urinary bladder, runs down and opens outside the body through urinary opening. It passes urine from bladder to outside the body.



**Fig. 2.10 Urinary System**

### 2.5.1 Structure of a Kidney

Kidney is enclosed in a membrane called peritoneum. A fluid is filled in between **peritoneum** and kidney called peritoneal fluid which reduces the friction. A longitudinal section of kidney shows three main parts: the cortex, the **medulla** and the **pelvis**.



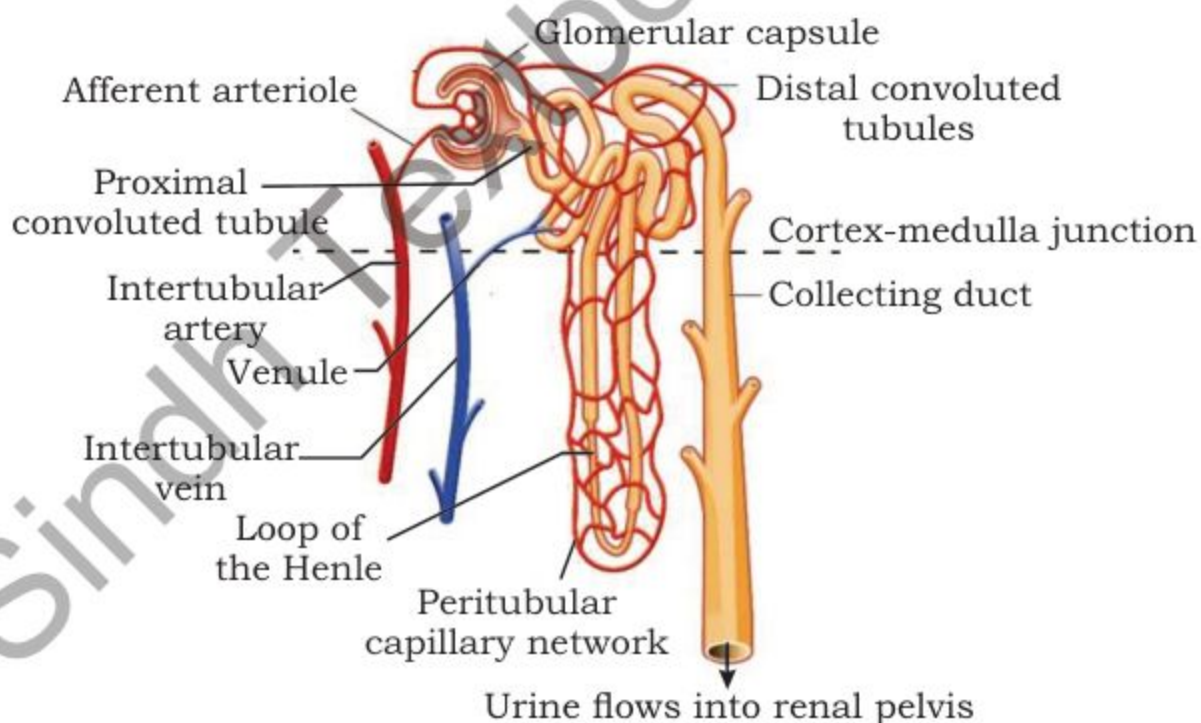
**Fig. 2.11 LS of Human Kidney**

**Cortex** is the outer dark brown portion. It is covered and protected by a fibrous capsule. The **medulla** is the inner lighter portion of the kidney. It contains the conical projection called **renal pyramids**; the human Kidneys contain 12-16 pyramids. The medulla contains nephrons. **Nephrons** are the basic functional units of the kidney. These are tiny kidney tubules where osmoregulation occurs to produce urine. The kidneys are connected to the ureter at pelvis. **Pelvis** is a funnel like space. It is the enlarged portion of ureter inside the kidney.

### 2.5.2 Structure of Nephron

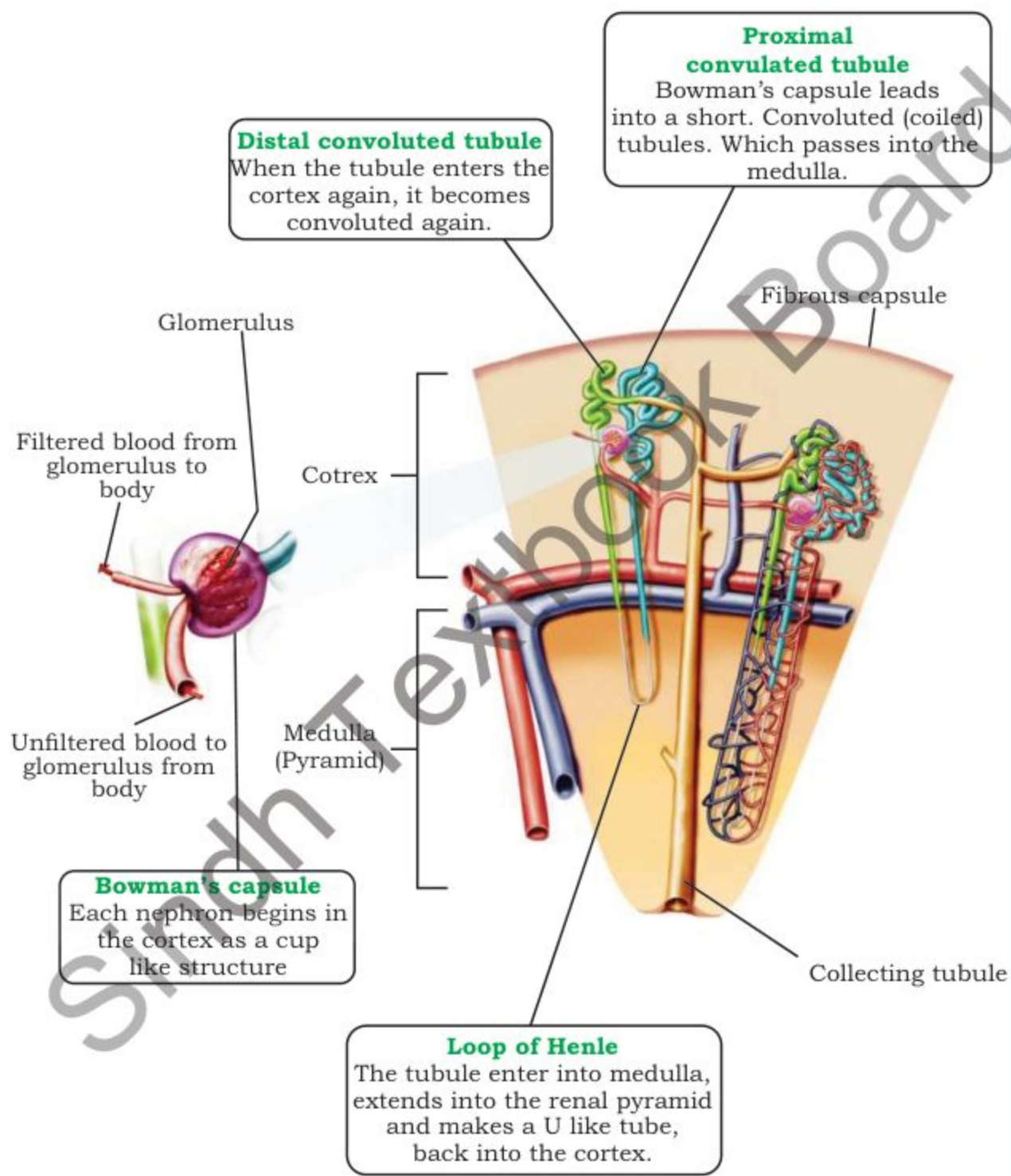
Nephrons are the functional unit of kidney. Each kidney contains more than one million nephrons, which are microscopic urinary tubules. Each nephron consists of four main parts: The Bowman's capsule, proximal convoluted tubule, Loop of Henle's, distal convoluted tubule. Number of nephrons open into a tube called **collecting duct**.

Nephrons are surrounded by different blood vessels that are connected to the renal artery and renal vein.



**Fig. 2.12 Structure of Nephron with Blood Vessels**





**Fig. 2.13 Section of Kidney showing structure of two nephrons with blood supply**

### 2.5.3. Function of Nephron

The renal artery when enters into kidney, it divides into millions of branches called afferent arteriole. The blood enters the kidney by renal artery goes into afferent arterioles.



Each afferent arteriole further divides into numerous blood capillaries in Bowmann's capsule are collectively called glomerulus. Bowmann's capsule with glomerulus are collectively called Malpighian body or renal corpuscle.



Blood leaving the glomerulus through efferent arteriole, enters blood capillaries surrounding the nephron.



Blood capillaries surround a loop of Henle's unite to form venule, which ultimately joins to form a branch of renal vein.

### 2.5.4. Role of kidney in urine formation

#### Urea formation

The urea is formed within the liver cells. The liver stores surplus glucose of food by converting it into glycogen and other food substances but it can not store the proteins. The excessive amino acid break and get



some energy from it. The amino group ( $\text{NH}_2$ ) is removed from amino acid called deamination. This  $\text{NH}_2$  group is converted into ammonia ( $\text{NH}_3$ ) which is very poisonous, it may kill the cell when stored in high concentration. So the liver cells quickly convert  $\text{NH}_3$  into less toxic substance **urea**. This urea is carried by blood to kidneys and excrete out in the form of urine. A small amount of urea is also excreted in sweat as well.

## Urine formation

Excess mineral salt and nitrogenous waste products i.e. urea, creatinine and uric acid, which are poisonous if accumulated. These are removed from body with water and this mixture is called urine.

Urine formation takes place in Kidneys. Two main processes are involved in the formation of urine within nephron.

- (i) Filtration
- (ii) Reabsorption

### (i) Filtration

Filtration is the process of taking out material from blood. It is of two types:

- (a) Ultrafiltration
- (b) Selective filtration

#### (a) Ultrafiltration

Ultrafiltration occurs at Malpighian body when the blood from afferent arteriole enters into glomerulus located in Bowman's capsule. Most of the blood plasma is forced out of the glomerulus blood capillaries into Bowman's capsule without any selection. Process of non-selective filtration is called ultrafiltration.

#### (b) Selective filtration

Selective filtration occurs at proximal and distal convoluted tubules when blood flows into peritubular capillaries, the remaining amount of urea filter out from blood by active transport. It requires some energy

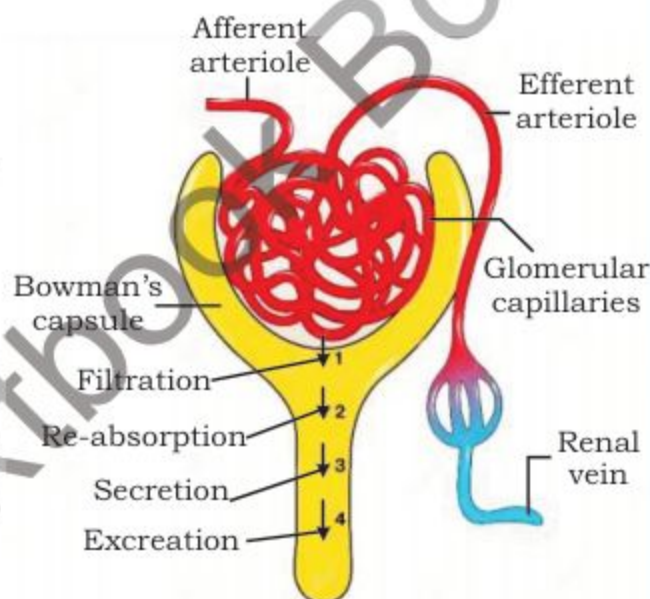


Fig. 2.14 Ultra Filtration in Glomerulus



## (ii) Reabsorption

In a normal adult about  $120 \text{ Cm}^3$  of filtrate is formed in the kidney every minute. If this large amount of filtrate allowed to pass out from the body as urine, the body will dehydrate and death may occur. To prevent this huge loss of water and useful salts, when the filtrate passes through the nephron useful substances and excessive water reabsorbed into the blood stream by:

- (a) Non-selective
- (b) Selective reabsorption

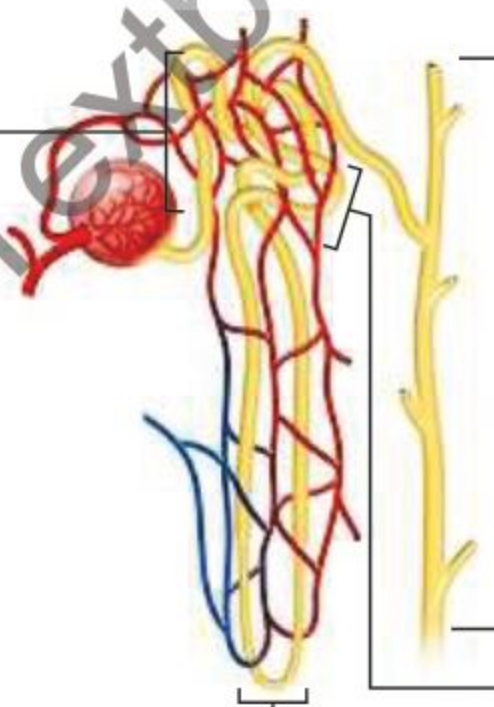
### (a) Non-selective reabsorption

Non-selective reabsorption occurs at distal and proximal convoluted tubules without any selection.

### (b) Selective reabsorption

Selective reabsorption occurs at Loop of Henle's and collecting duct with the help of hormones. i.e. Antidiuretic hormone (ADH), Parathyroid hormones (PTH) and calcitonin.

At proximal convoluted tubule, most of the mineral salts and in a healthy person, all of the glucose and amino acids are reabsorbed through the walls of the tubule into the surrounding blood capillaries. These solutes are reabsorbed via diffusion and active transport. This reabsorption is highly selective, and only those substances required by the body are reabsorbed readily. Most of the water in the filtrate is reabsorbed by osmosis here.



At the collecting duct, some water is reabsorbed. Excess water, excess salts and metabolic waste products such as urea, uric acid and creatinine pass out of the collecting duct into the renal pelvis as a mixture called urine.

At the distal convoluted tubule, some water and mineral salts are reabsorbed.

At the loop of Henle, some water is reabsorbed

**Fig. 2.15 Resorption of Material in Nephron**

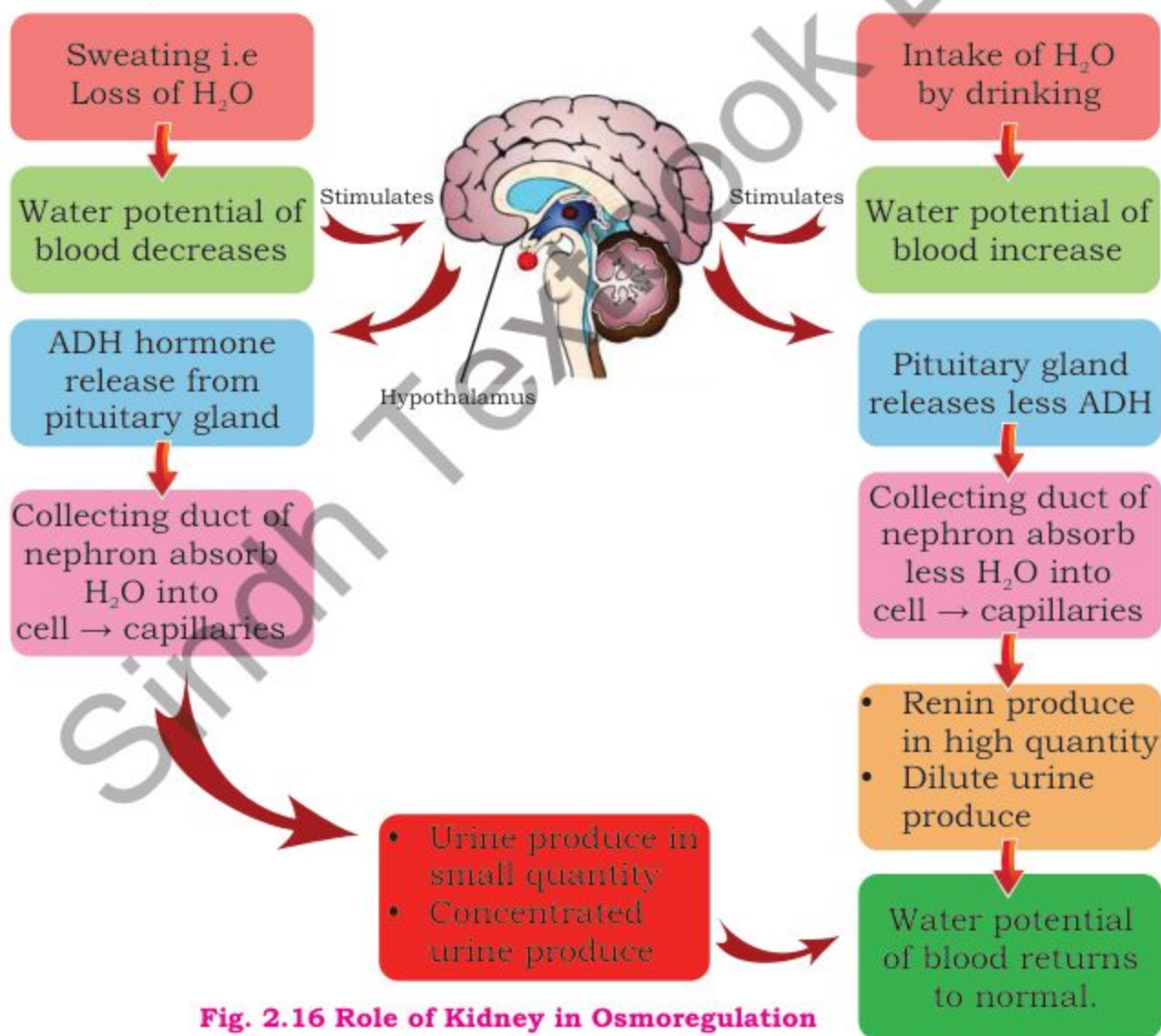


### 2.5.5. Role of kidneys in osmoregulation

The water potential (capacity to loose water) of blood in the body has to be kept constant because big and sudden change in the water potential of blood can lead to serious problems e.g. if plasma becomes very much dilute water will enter the blood cells will swell and possibly burst.

On the other hand if the blood plasma becomes too concentrated, water will move out of the cell by osmosis, as a result of it the blood cells, tissues will become dehydrated and shrink. This control of water and salt content of the body is known as osmoregulation.

Kidney is not only an excretory organ, it also regulates water and salt balance in the blood. Kidneys makes sure that the concentration of blood stays more or less constant.



**Fig. 2.16 Role of Kidney in Osmoregulation**

## 2.6 DISORDERS OF KIDNEYS

### Kidney Stone

A kidney stone is a solid mass that forms from the crystals of calcium oxalate or Calcium Carbonate. Sometimes uric acid and cysteine are also present in it. These molecules separate from urine, precipitate in kidney and deposit in the form of stone. Sometimes these stones are not hard therefore they break into sand like crystals which can pass out of the body with urine without pain. The little large size stone however damages the kidney tissues, it may stuck anywhere in urinary tract and cause renal failure with pain.



Fig. 2.17 Lithotripsy machine

### Treatment of Kidney Stone

If the size of stone is comparatively small we can use the technique of lithotripsy to break stone by ultrasonic waves (sound waves). The broken rudiments drain out from kidney with urine.

The large size stone cannot be broken by lithotripsy, so it is removed only by the process of renal surgery.

The large intake of water is the only measure to minimize the chances of formation of stone in kidney.

### Kidney Failure

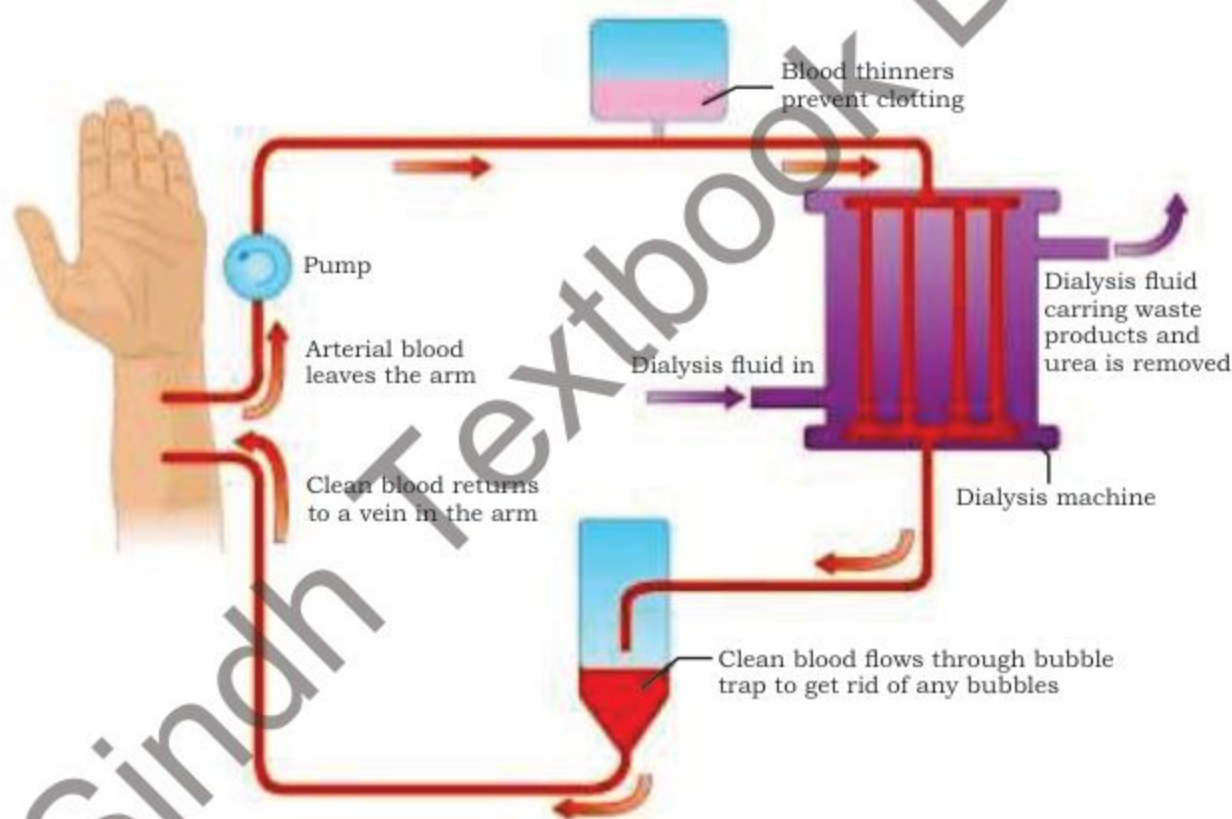
Sometimes the nephrons of kidney stop working, it is called kidney failure. It is mainly due to solute disbalance in blood and kidneys. The failure of kidneys allows urea and other waste material to accumulate in blood. The amount of  $H_2O$  is not regulated also. This disbalance of solutes causes death unless the patient is given treatment to filter out waste by machine.



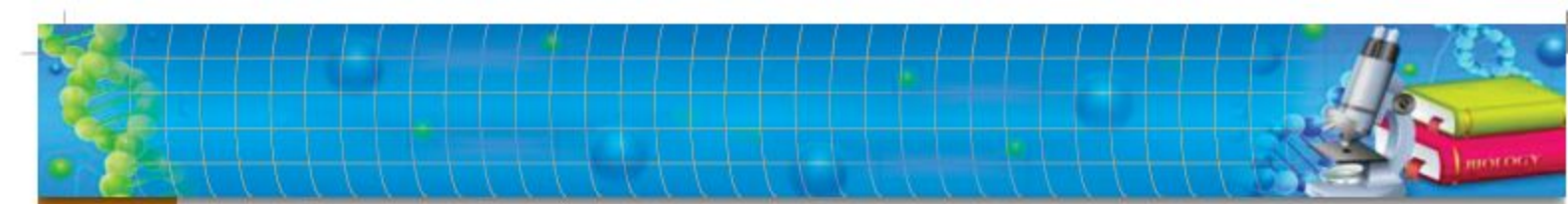
### (i) Kidney dialysis

Patient of kidney failure may get a kidney transplant. A person with two healthy kidneys may donate one kidney and survive with one kidney. If a donor is not available, the patient can be treated with dialysis using a dialysis machine. A dialysis machine performs the function of a kidney. It helps to clean the patient's blood from metabolic waste products and toxics.

For effective treatment the patient needs to undergo dialysis 2-3 times a week. Each session lasts about 3-5 hours depending on the patient's body size and medical condition.



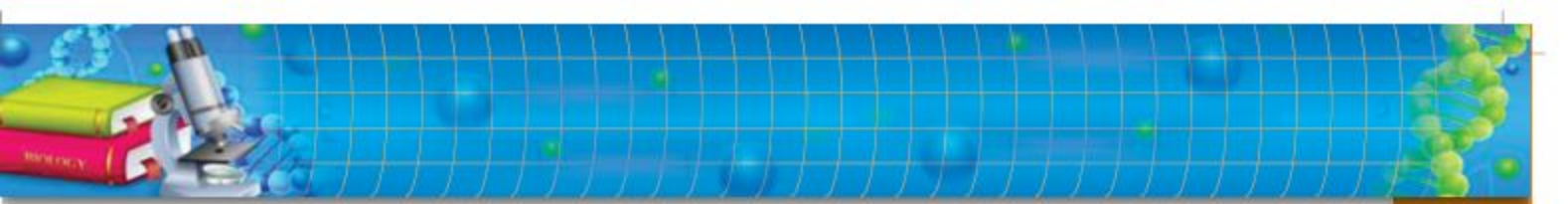
**Fig. 2.18 Working of Dialysis Machine**



## SUMMARY

- ◆ Homeostasis is set of metabolism which maintain internal environment of an organism within suitable limits.
- ◆ Organisms maintain internal condition by feedback mechanism.
- ◆ There are three main aspects of homeostasis i.e. osmoregulation, thermoregulation and excretion.
- ◆ Osmoregulation is the maintenance of internal water and salt condition by osmosis.
- ◆ Thermoregulation is the maintenance of temperature within suitable limits where enzymes can work optimumly.
- ◆ Excretion is the process where metabolic toxic wastes, excess substances remove from body.
- ◆ Hydrophytes are the plants which grow in fresh water.
- ◆ Halophytes grow in sea marshes or in saltish water.
- ◆ Mesophytes are the plants grow in moderate water containing soil.
- ◆ Xerophytes are the plants grow in soil of low water quantity.
- ◆ The main organs involved in human homeostasis are skin, lungs and kidneys.
- ◆ Human skin is consist of three layers called epidermis, dermis and hypodermis.
- ◆ Body temperature when rise body produce sweat, hair lies flat, vasodilation.
- ◆ When body temperature decrease the erection of hairs, vasocontraction, decrease in sweat production and increase in metabolic rate occurs.
- ◆ Lungs keep the  $\text{CO}_2$  concentration at low level.
- ◆ Kidneys maintain blood composition by maintain level of  $\text{H}_2\text{O}$ , solutes and nitrogenous waste by the process of filtration and reabsorption.
- ◆ Kidneys are situated at the dorsal side of the abdominal cavity on either side of the vertebral column.
- ◆ Nephrons are functional unit of kidney.
- ◆ Urea is formed in liver and excreted from kidneys.
- ◆ Mixture of urea, creatinine, uric acid and water called urine.
- ◆ Kidney stone is a solid mass made of calcium oxalate or carbonate crystals.
- ◆ Kidney failure patient can be treated with dialysis machine, dialysis machine performs the function of a kidney.





## EXERCISE

### A. MULTIPLE CHOICE QUESTIONS

**Choose the correct answer:**

- i) The internal condition of an organism is referred as:  
(a) Homeostasis (b) Internal environment  
(c) Internal metabolism (d) Feedback mechanism
- ii) A set of metabolism reaction which maintain internal environment is:  
(a) Positive feedback (b) Negative feedback  
(c) Osmoregulation (d) Homeostasis
- iii) Removal of extra liquid water is:  
(a) Exudation (b) Guttation  
(c) Respiration (d) Transpiration
- iv) Plant grow near coastal area called:  
(a) Xerophyte (b) Halophyte  
(c) Epiphyte (d) Hygrophyte
- v) Organ of human body which is considered on the largest organ is:  
(a) Skin (b) Digestive tract  
(c) Liver (d) Brain
- vi) The maintenance of body temperature with in suitable limit is called  
(a) Homeotherm (b) Thermoregulation  
(c) Osmoregulation (d) Heterotherm
- vii) The kidney is enclosed in a membrane called  
(a) Pericardium (b) Peritoneum  
(c) Pleural membrane (d) Plumule
- viii) The network of blood capillaries present in the layer of skin:  
(a) Epidermis (b) Dermis  
(c) Hypodermis (d) Endodermis
- ix) Selective reabsorption in nephron takes place at  
(a) Glomerulus (b) Malpighian body  
(c) Convulated tubules (d) Loop of Henle's
- x) The hormone ADH release from  
(a) Pituitary gland (b) Kidneys  
(c) Liver (d) Lungs

## **B. SHORT QUESTIONS:**

- i) Why homeostasis is required?
- ii) Why plants remove liquid water instead of water vapours?
- iii) How plant survive in saltish water?
- iv) Why skin is considered as excretory organ?
- v) What type of structures are present in dermis to perform different functions?
- vi) Draw a neat and labelled diagram of nephron.
- vii) What is kidney stone?
- viii) How human skin maintains temperature in cold conditions?
- ix) When dialysis is required?
- x) Why filtration at Para tabular capillaries called ultra filtration?

## **C. EXTENSIVE RESPONSE QUESTIONS:**

- i) How skin works as thermoregulatory organ?
- ii) Describe the urinary system of man with the help of diagram
- iii) Describe the structure of nephron within the L.S of kidney.
- iv) Describe the network of blood vesseles in nephron and their functions.
- v) Describe different disorders of kidneys and their treatment.



## Chapter 3

# COORDINATION

### Major Concept

In this Unit you will learn:

- Introduction
- Types of Coordination (Nervous and Chemical Coordination)
- Human Nervous System
  - ★ Components of Nervous System
  - ★ Structure and Function of Neuron
  - ★ Reflex Action and Reflex Arc
- Receptors of Man (Eyes and Ears)
- Endocrine System
  - ★ Important Endocrine Glands (Pituitary, Thyroid, Pancreas, Adrenal, Gonads)
- Nervous Disorders (Paralysis and Epilepsy)

